United International University

B.Sc. in Computer Science and Engineering (CSE)

CSE 4889: Machine Learning

Final Exam: Summer 2025   Time: 2 Hours   Marks: 40

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Answer all of the following questions **sequentially**.

1. a) Analyze the stochastic model of a neuron where Pseudo Temperature *T* is 0 and 1010, and the features of a single-layer neural network are as follows: [5]

Input	Weight
$I_1=2$	$W_1=0.3$
$I_2=4$	$W_2=0.5$
$I_3=3$	$W_3=0.1$

b) Consider the following training dataset: [5]

Input (I_1)	Input (I_2)	Desired Output (d)
0	0	0
1	1	1
0	1	1
Input (H)	Input (O)	------
0.1	0.2	

The initial weights and biases for the hidden and output layers are given as follows:

For the hidden layer	For the output layer
$W_{OH}=0.2$	$W_{HO}=0.3$
$W_{HH}=0.1$	$W_{OO}=0.2$
$W_{1H}=-0.2$	$W_{1O}=-0.3$
$W_{2H}=0.2$	$W_{2O}=0.3$
Bias $b_H=0.5$	Bias $b_O=0.4$

Assume:

- Learning rate $\eta=0.01$
- Momentum coefficient $\alpha=0.57$

Draw a schematic diagram of the Recurrent Neural Network showing all connections, including input-to-hidden, hidden-to-output, and recurrent links in the hidden and output layers.

2. You are working with a research team on a semi-supervised learning project. The dataset includes both labeled and unlabeled samples as follows: **[3+5+2]**

Type	Data Point (p)	Label (q)
Labeled	p_1=[3,2]	q_1=2
Labeled	p_2=[-2,-1]	q_2=1
Labeled	p_3=[2,3]	q_3=3
Unlabeled	p_4=[1,0]	-------
Unlabeled	p_5=[4,2]	-------
Unlabeled	p_6=[-3,1]	--------

The initial model parameters are:

- Weight vector: $\mathbf{w} = [0.40, -0.20]$
- Bias: $b = 0.5$
- Learning rate: $\eta = 0.25$
- Confidence threshold: $\tau = 0.65$

(i) Identify which unlabeled samples qualify for pseudo-labeling based on τ.
(ii) Using both labeled and pseudo-labeled samples, compute the updated weights (w_{new}) and bias (b_{new}) after one iteration of learning.
(iii) Interpret the result and discuss how the confidence threshold influences learning.

3. Consider the following sentence fragment for next-word prediction: [1+5+4]

"We Love My"

Assume that the model consists of one self-attention layer with a vector dimension of 2, and the attention dimension is $d_k = 3$. The vocabulary for possible next tokens is:

{**Mother, Teacher, Father, Brother**}

For this task, use the self-attention mechanism with parameters defined as:

Q=K=V=X

(i) Construct the input matrix X for the tokens in the sentence fragment.
(ii) Derive the final representation for the sequence after the self-attention layer.
(iii) Based on the resulting context vector, predict the most probable next token from the given vocabulary.

4. Consider the following dataset representing input–output pairs: [8+2]

Input (X)	Output (Y)
(5,1)	6
(4,5)	5
(7,3)	9

A General Regression Neural Network is to be used to predict the output corresponding to a new input value of X = (6, 6).

(i) Compute the predicted output $\hat{Y}(6)$ using the following smoothing parameters:

smoothing parameters σ
1
0.6
0.4

(ii) Discuss how the choice of σ affects the smoothness and generalization of the GRNN model.